

Yuktha Priya Masupalli

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EDUCATION

- Texas A&M University-San Antonio** San Antonio, USA
MS in Computer Science; GPA: 3.87/4.0 Aug 2024 - Dec 2025
- JNTUA College of Engineering** Kalikiri, India
B.Tech in Computer Science; GPA: 8.1/10 Nov 2020 - May 2023
- S.V Govt. Polytechnic** Tirupati, India
Diploma in Computer Engineering; Percentage: 93.6/100 Jun 2017 - Apr 2020

SKILLS SUMMARY

- Languages:** Python, Java, JavaScript, PHP, HTML/CSS, SQL, C, C++, Kotlin, C, Scala, Rust
- Frameworks:** ReactJS, AngularJS, NodeJS, Django, TensorFlow, Pytorch, Keras
- Tools and Platforms:** AWS, GCP, Azure, PostgreSQL, MySQL, SQLite, MongoDB, Neo4j, Kubernetes, Git
- Soft Skills:** Leadership, Event Management, Adaptability, Public Speaking, Problem-Solving, Time Management
- Distributed Systems:** Familiar with Hadoop, SAS, Big Data Analytics, HPC
- Certification:** Microsoft Certified: Azure AI Fundamentals view certificate

EXPERIENCE

- Texas A&M University-San Antonio** San Antonio, TX
Graduate Research Assistant Feb 2025 - May 2025
 - Designed and implemented a scalable **cross-domain information retrieval system** using CLIP, retrieving semantically relevant captions for query images from a dataset of over 100,000 entries.
 - Optimized data pipelines and leveraged parallel processing techniques, reducing retrieval latency by 20% on large-scale multimodal datasets.
 - Developed improved probabilistic image-text representations for **image-text matching**, boosting performance on **COCO** and **CXR-CLIP** datasets through advanced embedding and optimization techniques.
 - Researched and fine-tuned **diffusion models** and large language models (LLMs) for cross-modal content generation, integrating vision-language embeddings to enhance semantic understanding and improve image captioning and multimodal alignment accuracy.
 - Utilized **CUDA-based acceleration** in HPC environments to train and evaluate multiple models efficiently.
 - Contributed to the **JagCoach AI assistant** project by developing audio-video analysis pipelines for feedback generation on oral presentations and managing machine learning model deployment and resource optimization on Google Cloud Platform (GCP).
- Digitalsoft AI Solutions India Pvt. Ltd(DAIS)** Chennai, India
Associate Data Engineer July 2023 - Jun 2024
 - Orchestrated UI design using AngularJS, elevating project frontend for a superior user interface.
 - Enhanced project functionality by integrating APIs, optimizing scalability and reliability through AWS services.
 - Pioneered research on recommendation systems, refining user experiences through innovative solutions.
 - Developed GenAI-powered chat assistants using LangChain and LLMs, enabling intelligent user interactions.

PROJECTS

- Rust System Programming Vulnerabilities Detection Using Machine Learning and Static Analysis Tools:**
 - Developed a system to detect vulnerabilities in Rust programming using machine learning and static analysis.
 - Utilized Rust's strong type system and borrow checker to minimize memory safety issues.
 - Applied machine learning models to classify potential security vulnerabilities with high accuracy.
 - Integrated static analysis tools to automate vulnerability detection, improving software security.
- University Recommendation System (Neo4j, Machine Learning):**
 - Used Neo4j's graph database and KNN algorithm to generate personalized university recommendations.
 - Calculated similarity scores based on GPA, rankings, and reputation in engineering and business departments.
 - Developed a Flask web app, integrating Neo4j queries for real-time user input and personalized suggestions.
- Public HealthCare Monitoring and Diabetes Prediction System (NLP, Classification Algorithms):**
 - Developed a web application for public healthcare, focusing on diabetes prediction with NLP and classification algorithms.
 - Created an interactive and responsive UI with HTML, CSS, and JavaScript to improve user experience.
 - Integrated a SQLite database for secure user data storage, enabling seamless data retrieval and updates.

PUBLICATIONS

- Benchmarking the Robustness of Contrastive Learning Models for Medical Image-Report Retrieval under Occlusion Attacks** – Accepted at AAAI Workshop. [Link](#)
- Improving Medical Imaging Model Calibration through Probabilistic Embedding** – Published at IEEE Big Data Conference. [Link](#)